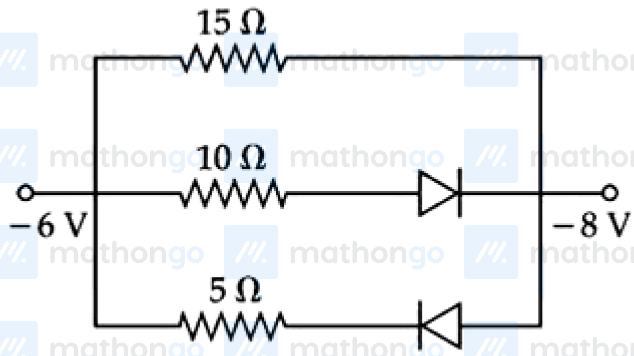
Q1 - 2024 (04 Apr Shift 1)

The value of net resistance of the network as shown in the given figure is :



(1) 6Ω

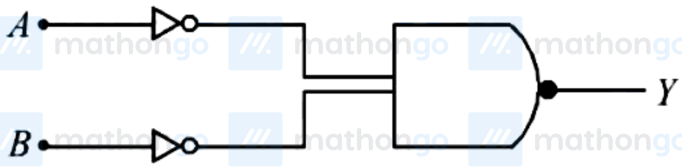
(2) $(5/2)\Omega$

(3) $(15/4)\Omega$

(4) $(30/11)\Omega$

Q2 - 2024 (04 Apr Shift 2)

Identify the logic gate given in the circuit:



(1) NAND-gate

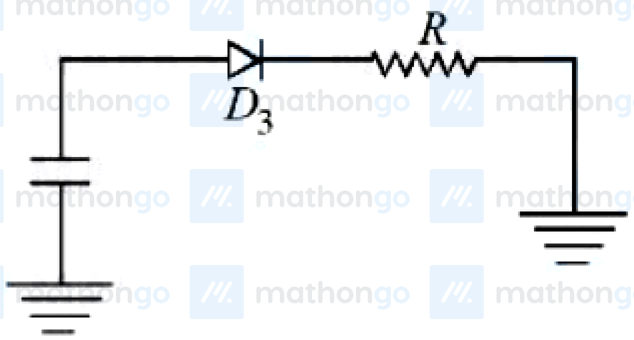
(2) AND gate

(3) NOR gate

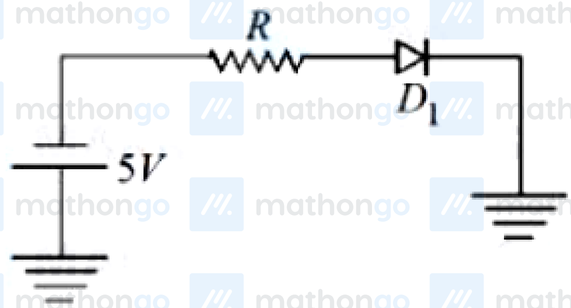
(4) OR- gate

Q3 - 2024 (04 Apr Shift 2)

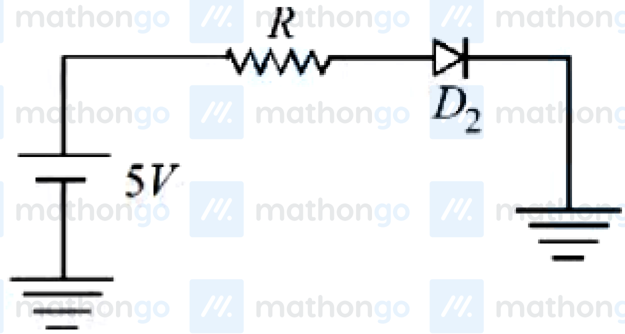
Which of the diode circuit shows correct biasing used for the measurement of dynamic resistance of p-n junction diode :



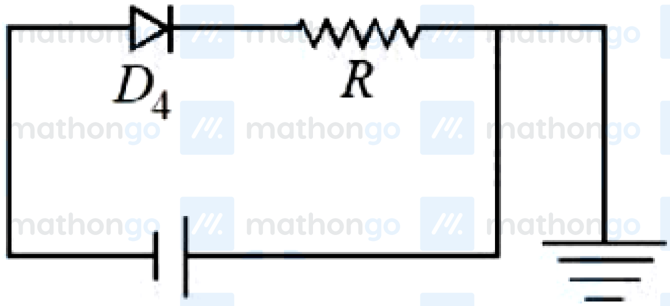
(1)



(2)



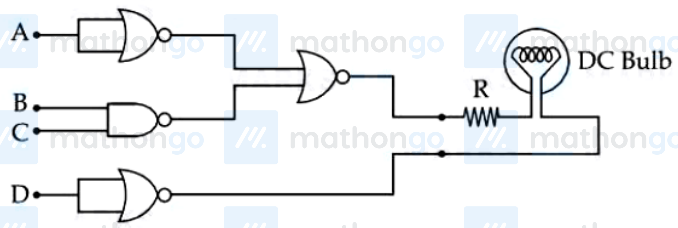
(3)



(4)

Q4 - 2024 (05 Apr Shift 1)

Following gates section is connected in a complete suitable circuit.



For which of the following combination, bulb will glow (ON) :

(1) $\mathrm{A}=0,\ \mathrm{B}=0, \mathrm{C}=0, \mathrm{D}=1$

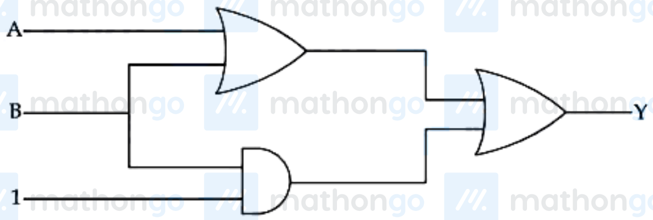
(2) $\mathrm{A}=0,\ \mathrm{B}=1, \mathrm{C}=1, \mathrm{D}=1$

(3) $\mathrm{A}=1,\ \mathrm{B}=0, \mathrm{C}=0, \mathrm{D}=0$

(4) $\mathrm{A}=1,\ \mathrm{B}=1, \mathrm{C}=1, \mathrm{D}=0$

Q5 - 2024 (05 Apr Shift 2)

The output (Y) of logic circuit given below is 0 only when :



(1) $\mathrm{A}=1,\ \mathrm{B}=0$

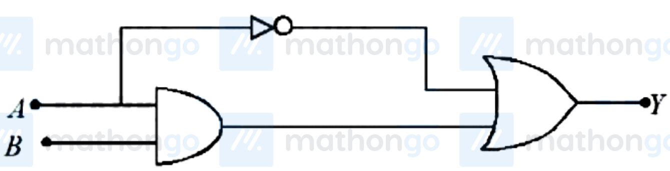
(2) $\mathrm{A}=0,\ \mathrm{B}=1$

(3) $\mathrm{A}=0,\ \mathrm{B}=0$

(4) $\mathrm{A}=1,\ \mathrm{B}=1$

Q6 - 2024 (06 Apr Shift 1)

The correct truth table for the following logic circuit is :



(1)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

(2)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(3)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(4)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Q7 - 2024 (06 Apr Shift 2)

The acceptor level of a p-type semiconductor is 6eV. The maximum wavelength of light which can create a hole would be : Given $hc = 1242 \mathrm{eVnm}$.

(1) 414 nm

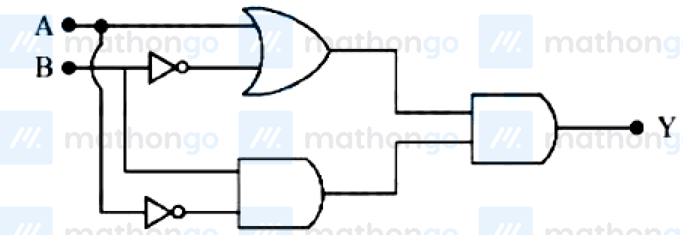
(2) 103.5 nm

(3) 207 nm

(4) 407 nm

Q8 - 2024 (08 Apr Shift 1)

The output Y of following circuit for given inputs is :



(1) $\Lambda \cdot B(\Lambda + B)$

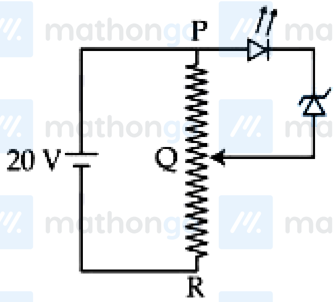
(2) 0

(3) $\overline{A} \cdot B$

(4) $A \cdot B$

Q9 - 2024 (08 Apr Shift 2)

A potential divider circuit is connected with a dc source of 20 V, a light emitting diode of glow in voltage 1.8 V and a zener diode of breakdown voltage of 3.2 V. The length (PR) of the resistive wire is 20 cm. The minimum length of PQ to just glow the LED is _____cm



Q10 - 2024 (09 Apr Shift 1)

A light emitting diode (LED) is fabricated using GaAs semiconducting material whose band gap is 1.42eV. The wavelength of light emitted from the LED is :

(1) 1400 nm

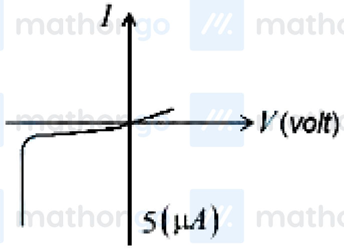
(2) 650 nm

(3) 875 nm

(4) 1243 nm

Q11 - 2024 (09 Apr Shift 2)

The $I-V$ characteristics of an electronic device shown in the figure. The device is:



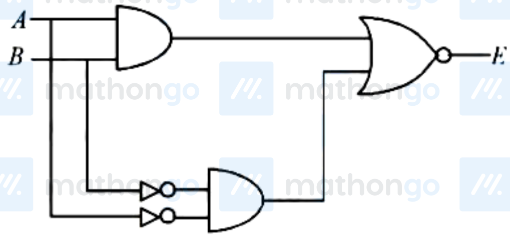
(1) a diode which can be used as a rectifier

(2) a zener diode which can be used as a voltage regulator

(3) a transistor which can be used as an amplifier

(4) a solar cell

Q12 - 2024 (09 Apr Shift 2)



A	B	E
0	0	0
0	1	X
1	0	Y
1	1	0

In the truth table of the above circuit the value of X and Y are :

(1) 0,0

(2) 1,1

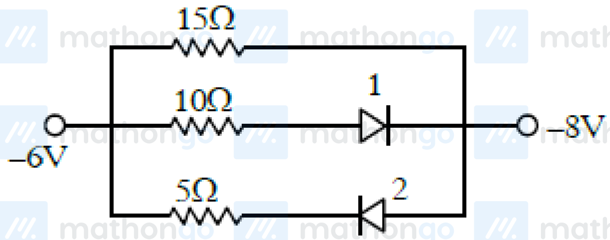
(3) 1,0

(4) 0,1

Answer Key

Q1 (1)	**Q2** (4)	**Q3** (3)	**Q4** (3)
Q5 (3)	**Q6** (1)	**Q7** (3)	**Q8** (2)
Q9 (5)	**Q10** (3)	**Q11** (2)	**Q12** (2)

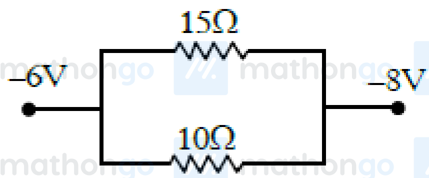
Q1



Diode 2 is in reverse bias

So current will not flow in branch of 2nd diode, So we can assume it to be broken wire.

Diode 1 is in forward bias



So it will behave like conducting wire. So new circuit will be

$$R_{eq} = \frac{15\times10}{15+10} = \frac{15\times10}{25} = 6\Omega$$

Q2

$Y = \overline{\overline{A}\cdot\overline{B}}$

By De-Morgan Law

$Y = \overline{\overline{A+B}}$

$Y = A + B$

Hence OR gate

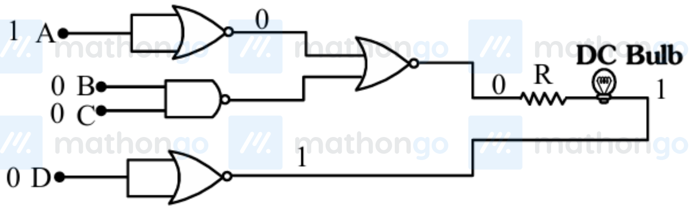
Q3

Diode should be in forward biased to calculate dynamic resistance.

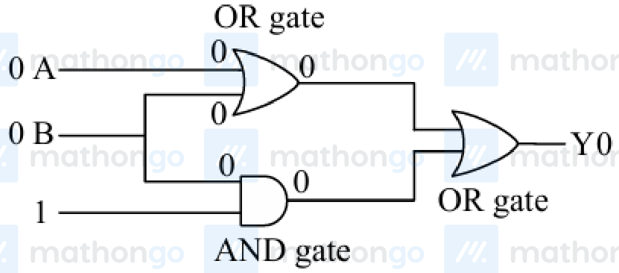
Q4

Bulb will glow if bulb have potential drop on it. One end of bulb must be at high (1) and other must be at low (0).

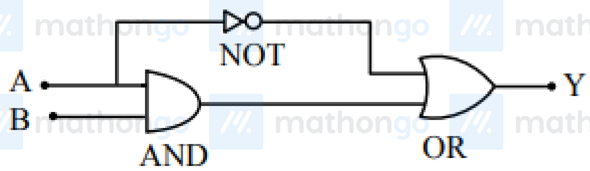
Option (3) satisfy this condition



Q5



Q6



Q7

$$\text{Energy} = \frac{hc}{\lambda}$$

$$E = \frac{1240}{\lambda(nm)} eV$$

$$6 = \frac{1240}{\lambda(nm)}$$

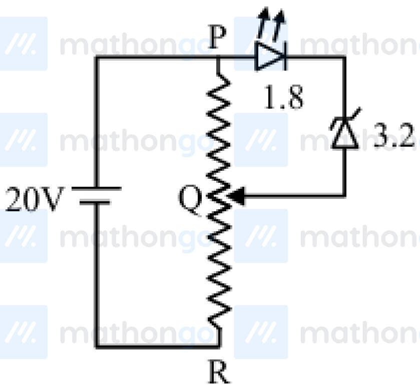
$$\lambda = \frac{1240}{6} = 207 \text{ nm}$$

Q8

By truth table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	0

Q9



$PR = 20 \text{ cm}$

$V_{PQ} = \frac{1}{4} \times R_{PR}$

$\ell_{\min}(PQ) = \frac{1}{4} \times 20$

$= 5 \text{ cm}$

Q10

$\lambda = \frac{1240}{1.42} = 875 \text{ nm (Approx)}$

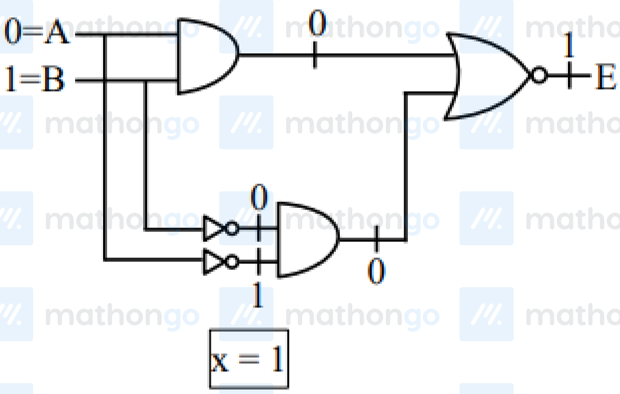
Q11

Theory

Zener diode used as voltage regulator

Q12

For x



For y